

6.3 Ecosystems

6.3.1 Ecosystems

Organisms do not live in isolation but engage in complex interactions, not just with other organisms but also with their environment.

The efficiency of biomass transfer limits the number of organisms that can exist in a particular ecosystem.

Ecosystems are dynamic and tend towards some form of climax community.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) ecosystems, which range in size, are dynamic and are influenced by both biotic and abiotic factors	To include reference to a variety of ecosystems of different sizes (e.g. a rock pool, a playing field, a large tree) and named examples of biotic and abiotic factors.

(b) biomass transfers through ecosystems

To include how biomass transfers between trophic levels can be measured

AND

the efficiency of biomass transfers between trophic levels

$$\text{efficiency} = \frac{\text{biomass transferred}}{\text{biomass intake}} \times 100$$

AND

how human activities can manipulate the transfer of biomass through ecosystems.

M0.1, M0.2, M0.3, M0.4, M1.1, M1.3, M1.6
HSW12

(c) recycling within ecosystems

To include the role of decomposers and the roles of microorganisms in recycling nitrogen within ecosystems (including *Nitrosomonas*, *Nitrobacter*, *Azotobacter* and *Rhizobium*)

AND

the importance of the carbon cycle to include the role of organisms (decomposition, respiration and photosynthesis) and physical and chemical effects in the cycling of carbon within ecosystems.

HSW2, HSW12

(d) the process of primary succession in the development of an ecosystem

To include succession from pioneer species to a climax community

AND

deflected succession.

HSW12

(e) **(i)** how the distribution and abundance of organisms in an ecosystem can be measured

M1.3, M1.4, M1.5, M1.7, M1.9, M1.10, M3.1, M3.2

PAG3

(ii) the use of sampling and recording methods to determine the distribution and abundance of organisms in a variety of ecosystems.

HSW4